

Methodology

Request and canonical artifact I

DuckRabbit starts from a typed request

$$q = (i, \theta, s, e)$$

{#eq:typed_request}

where i is an illusion identifier, θ is an immutable parameter object, s is a deterministic seed, and e is an optional encoding request. A registered generator maps the request to a canonical artifact:

$$A = G_i(\theta; s), \quad A \in \{I, X, V, AV\}$$

{#eq:canonical_generation}

Request and canonical artifact II

The four artifact types are an image frame I , audio buffer X , video sequence V , and audiovisual timeline AV . The generator core is independent of Pillow, WAV containers, GIF, MP4, and ffmpeg. Each artifact retains its shape, dtype, units, clock, channels, and timing offsets.

Typed parameter domains I

Value objects validate finite ranges at construction time. Image parameters include dimensions, color mode, luminance bounds, grayscale levels, and quantization levels. Audio parameters include frequency, phase, amplitude, duration, envelope, sample rate, channel count, and PCM depth. Video parameters include dimensions, rational frame rate, frame count, and temporal offsets. Audiovisual parameters compose audio and video clocks with declared synchronization and spatial discrepancies.

The canonical sample count and frame timestamps are explicit. For audio duration T and sample rate f_s , N is rounded once at construction. For a video with frame rate f_r , frame index k is timestamped on the presentation clock. The audio-minus-video sign convention is explicit:

$$N = \text{round}(f_s T), \quad t_k = k / f_r, \quad \Delta_{AV} = t_{\text{audio}} - t_{\text{video}}$$

Typed parameter domains II

`{#eq:clock_definition}`

The package rejects non-finite values, invalid shapes, out-of-range samples, aliasing frequencies, incompatible channel layouts, impossible frame rates, and contradictory encoding requests.

Canonical serialization and provenance I

Canonical arrays are contiguous little-endian float32 buffers with explicit metadata. Serialization includes the array values, shape, clock, and units; metadata are part of the identity rather than an informal annotation:

$$c(A) = \text{Serialize}_{\text{LE}, \text{float32}}(A, \text{shape}, \text{clock}, \text{units}), \quad h_c(A) = H(c(A))$$

{#eq:canonical_digest}

The canonical digest is independent of PNG, WAV, GIF, MP4, and NPZ delivery choices. The v2 manifest records the parameter schema, serialized parameters, taxonomy, evidence boundary, canonical facts, objective metrics, encoding profile, backend identity, decoded inspection, and verification status. A v1 reader remains available and marks upconverted records unverified until the encoded file is re-inspected.

Canonical serialization and provenance II

This separation follows a reproducibility principle from computational research: a result is not made reproducible merely by publishing a final file; the executable inputs, transformation steps, versioned software, and provenance must remain identifiable [@sandve2013simple; @wilson2014bestpractices]. The manifest therefore records the construction identity and delivery identity separately. A downstream user can cite the software version and regenerate the canonical object, or cite a particular encoded artifact when the delivery file itself is the relevant research object [@smith2016software].

Canonical serialization and provenance III

The distinction also follows the older psychophysical problem of relating a controlled physical increment to a measured sensation. Fechner's 1860 treatise made measurement, stimulus control, and response comparison explicit parts of the scientific object [fechner 1860psychophysics]. DuckRabbit adopts the stimulus-control and provenance aspects of that tradition, but it does not pretend that a canonical digest or media metric is a sensation measurement: observer data still require a task, presentation conditions, and a registered analysis.

Encoding and verification I

For an encoding request e , the delivered file F and decoded inspection I are:

$$F = E_e(A), \quad I = D(F), \quad V(F, M) \in \{\text{pass}, \text{fail}\}$$

{#eq:encoding_verification}

The manifest M binds A , F , and I through hashes, typed summaries, and format-specific tolerances. PNG, GIF, WAV, and NPZ use local deterministic adapters. MP4 and muxed audiovisual output use an optional ffmpeg backend. Missing backends are reported as capability errors. Writes are atomic, parent directories are created, overwrite behavior is explicit, and a conflicting format argument is rejected.

Objective metric definitions I

MetricRecord and MetricSuite carry a metric name, value, unit, computation version, source artifact, tolerance, and claim level.

Luminance statistics, quantization levels, RMS and peak amplitude, spectral summaries, frame deltas, stream durations, and synchronization offsets are physical or decoded-media facts. They are not observer responses and cannot by themselves establish an illusion effect. For canonical image samples $y_1, \dots, y_n$, audio samples $x_1, \dots, x_n$, and adjacent video frames Y_k, Y_{k+1} , the principal metrics are:

$$\mu_Y = \frac{1}{n} \sum_{j=1}^n y_j, \quad \sigma_Y = \sqrt{\frac{1}{n} \sum_{j=1}^n (y_j - \mu_Y)^2}, \quad RMS_X = \sqrt{\frac{1}{n} \sum_{j=1}^n x_j^2}$$

{#eq:objective_statistics}

Objective metric definitions II

$$D_k = \frac{1}{|Y|} \sum_p |Y_{k+1}(p) - Y_k(p)|, \quad \text{centroid}(X) = \frac{\sum_f f P_X(f)}{\sum_f P_X(f)}$$

{#eq:temporal_spectral_metrics}

The level count is the cardinality of the finite-value set after canonical quantization; an empty or non-finite artifact is invalid rather than assigned a default statistic. Every metric record carries units and a deterministic computation version. Observer interpretations are represented separately by the typed observer protocol and claim levels.

For a future observer study, the estimand is defined by the protocol rather than by the generator itself:

$$\psi = \mathbb{E}[Y \mid \text{condition, protocol}]$$

Objective metric definitions III

`{#eq:observer_estimand}`

This symbol names a planned analysis target only. DuckRabbit ships no human responses, fitted coefficients, or validated observer-level estimate.

Objective metric definitions IV

Synthetic psychophysics boundary

DuckRabbit v0.5.0 adds a transparent synthetic-psychophysics diagnostic for method orchestration. It is a hand-specified feature observer, not a pretrained vision model and not an empirical psychophysics substitute. Let $\phi(A)$ be the bounded feature vector extracted from a canonical artifact, w the code-owned feature weights, and $\tau > 0$ the declared temperature:

$$p_M(c \mid A_r, A_c) = \sigma \left(\frac{w^\top \phi(A_c) - w^\top \phi(A_r)}{\tau} \right)$$

{#eq:synthetic_observer}

The model identity, version, preprocessing features, weights, temperature, seed, calibration statement, and the `human_data=false` flag are serialized with each prediction. The diagnostic is useful for checking that typed parameter sweeps produce deterministic, inspectable model outputs and for exercising analysis orchestration end to end. It does not estimate a human psychometric function, establish a sensitivity threshold, or support